GREEEN tallgrass prairie restoration: RLQ analyses timing x herbicide (Lux Arbor)

MB

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Preparation

```
library(here)
library(tidyverse)
library(ade4)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types =
    cols(
    .default = "?",
    seeding_time = "f"
  )) %>%
  arrange(id_plot_year) %>%
  filter(
    site == "Lux Arbor",
```

```r
    richness_type == "seeded_richness",
    treatment_id %in% c("1", "2", "3", "4")
  ) %>%
  mutate(
    treatment = str_c(seeding_time, herbicide, seeded_pool, sep = "_"),
    treatment = factor(treatment)
  ) %>%
  filter(treatment != "unseeded_0_0") %>%
  select(
    id_plot_year, year, treatment, water_cap, cover_non_seeded,
    cover_seeded_grass
    )

traits <- read_csv(
  here("data", "processed", "data_processed_traits.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(seeded == "1", family != "Poaceae") %>%
  select(accepted_name, sla, height, seedmass) %>%
  mutate(seedmass = log(seedmass)) %>%
  drop_na()

species <- read_csv(
  here("data", "processed", "data_processed_species.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(.default = "?")
  ) %>%
  filter(plot_size %in% c("1", "25")) %>%
  mutate(abundance = 1) %>%
  select(id_plot_year, accepted_name, abundance) %>%
  semi_join(sites, by = "id_plot_year") %>%
  semi_join(traits, by = "accepted_name") %>%
  arrange(id_plot_year)
```

Load data

Checks

```r
sites %>%
  group_by(year) %>%
  count()
```

Plots per year

```
## # A tibble: 6 x 2
## # Groups:   year [6]
##    year     n
##   <dbl> <int>
## 1  2015    17
## 2  2016    18
## 3  2017    18
## 4  2018    18
## 5  2019    18
## 6  2025    18
```

```r
species %>%
  group_by(id_plot_year) %>%
  count(name = "seeded_forb_richness") %>%
  ungroup() %>%
  count(seeded_forb_richness, name = "number_of_plots") %>%
  mutate(
    "plots_>=_1_species" = sum(number_of_plots),
    total_number_plots = 108
    )
```

Seeded species richness

```
## # A tibble: 16 x 4
##    seeded_forb_richness number_of_plots `plots_>=_1_species` total_number_plots
##                   <int>           <int>                <int>              <dbl>
##  1                    1               1                  104                108
##  2                    2               1                  104                108
##  3                    3               6                  104                108
##  4                    4               7                  104                108
##  5                    5               5                  104                108
##  6                    6              10                  104                108
##  7                    7              14                  104                108
##  8                    8              11                  104                108
##  9                    9              12                  104                108
## 10                   10              12                  104                108
## 11                   11               8                  104                108
## 12                   12              10                  104                108
## 13                   13               1                  104                108
## 14                   14               3                  104                108
## 15                   16               2                  104                108
## 16                   17               1                  104                108
```

```r
traits %>% anti_join(species, by = "accepted_name")
```

Check non-occuring seeded species

```
## # A tibble: 6 x 4
##   accepted_name                  sla height seedmass
##   <chr>                        <dbl>  <dbl>    <dbl>
## 1 Dalea purpurea                191.  0.91     -6.07
## 2 Liatris cylindracea           110.  0.564    -5.73
## 3 Lupinus perennis              167.  0.326    -3.84
## 4 Silphium terebinthinaceum     171.  0.716    -4.53
## 5 Solidago juncea               130.  0.783    -7.79
## 6 Symphyotrichum oolentangiense 143.  0.362    -8.68
```

Finalize preparation

```r
traits <- traits %>%
  semi_join(species, by = "accepted_name") %>%
  column_to_rownames(var = "accepted_name")

sites <- sites %>%
```

```r
  semi_join(species, by = "id_plot_year") %>%
  column_to_rownames(var = "id_plot_year")

species <- species %>%
  pivot_wider(
    names_from = "accepted_name", values_from = "abundance", values_fill = 0
    ) %>%
  column_to_rownames(var = "id_plot_year")
```

Statistics

Descriptive analyses

```r
sites %>%
  group_by(year) %>%
  count()
```

```
## # A tibble: 6 x 2
## # Groups:   year [6]
##    year     n
##   <dbl> <int>
## 1  2015    14
## 2  2016    18
## 3  2017    18
## 4  2018    18
## 5  2019    18
## 6  2025    18
```

Ordinations

```r
dudi_l <- dudi.coa(species, scannf = FALSE)
dudi_r <- dudi.hillsmith(
  sites,
  row.w = dudi_l$lw,
  scannf = FALSE
  )
dudi_q <- dudi.pca(
  traits,
  row.w = dudi_l$cw,
  scannf = FALSE
  )
```

RLQ analyses

```r
m_rlq <- rlq(
  dudiR = dudi_r,
  dudiL = dudi_l,
  dudiQ = dudi_q,
  scannf = FALSE
  )
```

Global test

Model 2: species-environment relationship

Model 4: species-traits relationship

```
randtest(m_rlq, nrepet = 999, modeltype = 6)
```

```
## class: krandtest lightkrandtest
## Monte-Carlo tests
## Call: randtest.rlq(xtest = m_rlq, nrepet = 999, modeltype = 6)
##
## Number of tests:   2
##
## Adjustment method for multiple comparisons:   none
## Permutation number:   999
##      Test       Obs   Std.Obs   Alter Pvalue
## 1 Model 2 0.1130885 19.159409 greater  0.001
## 2 Model 4 0.1130885  1.585155 greater  0.081
```

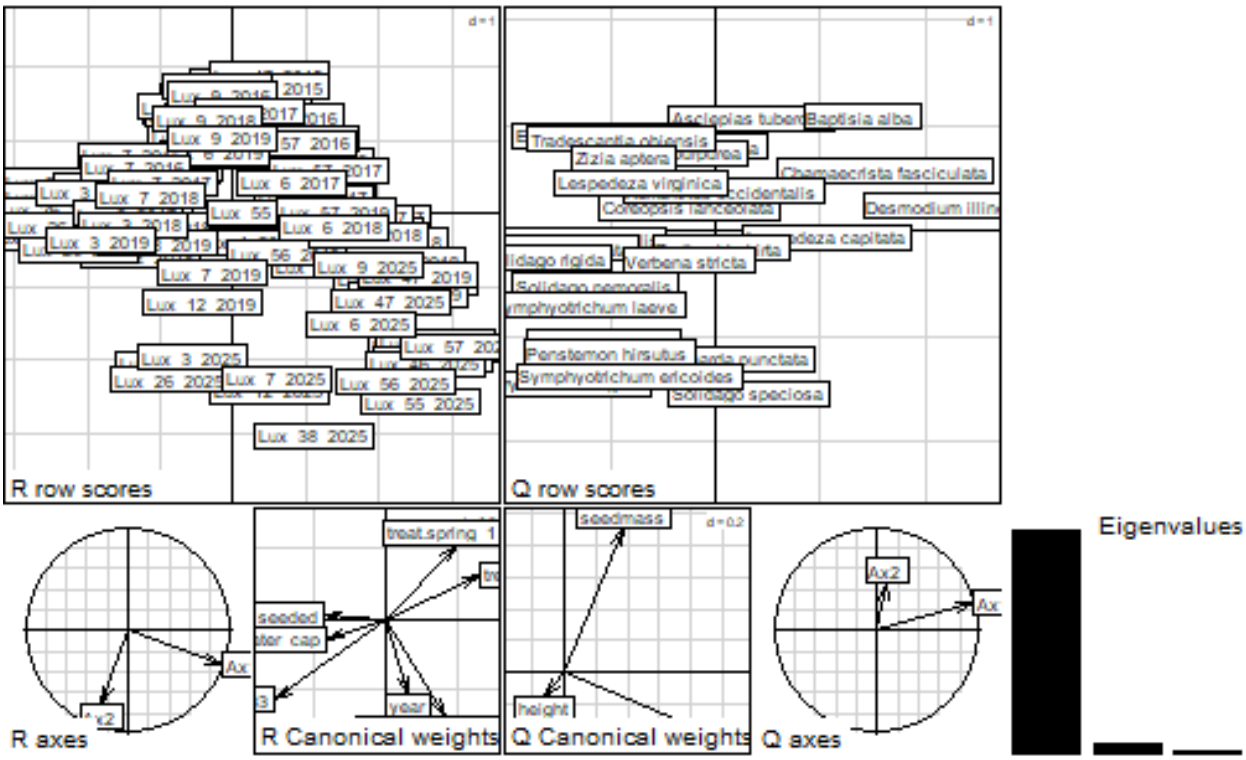
Traits by environment crossed table

```
m_rlq$tab
```

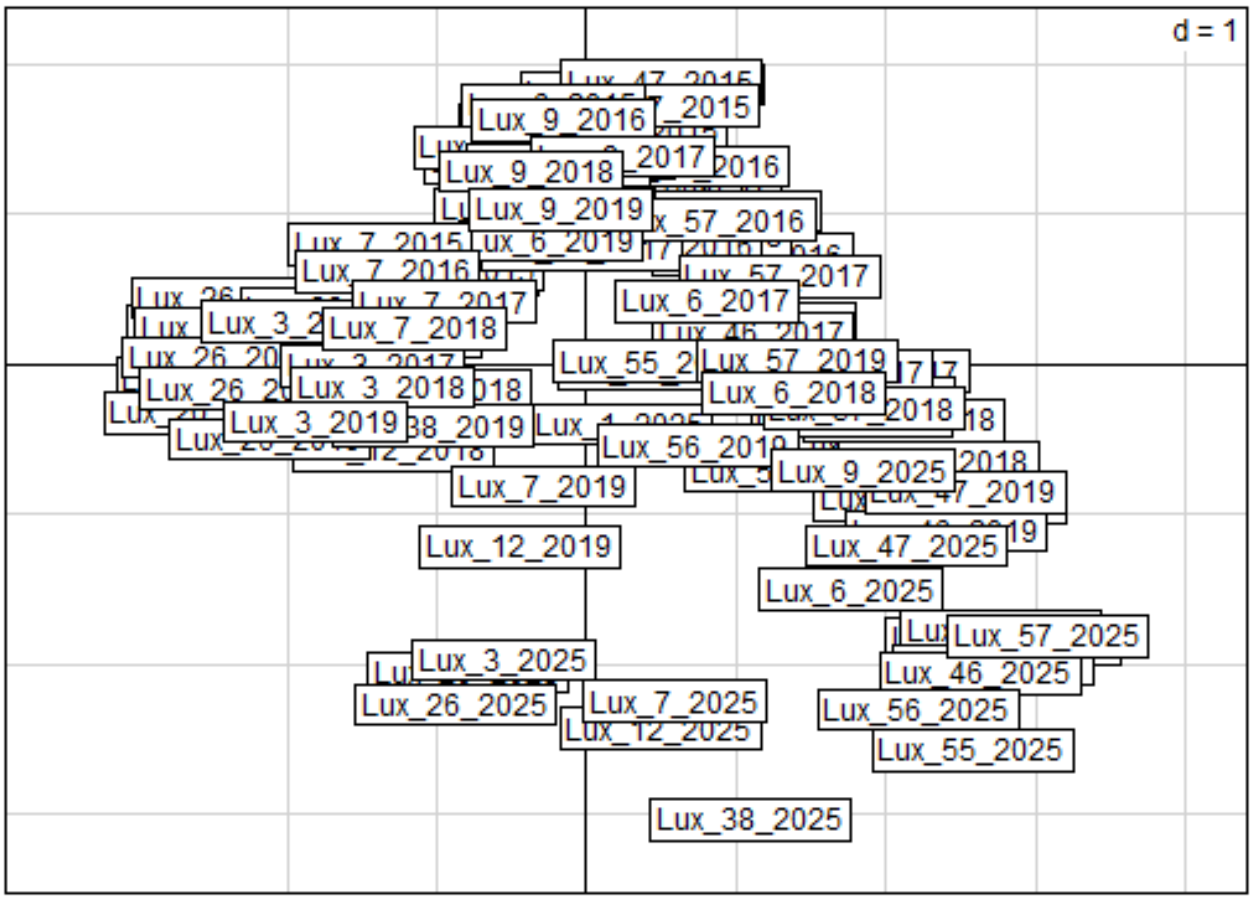
```
##                            sla       height     seedmass
## year                 0.06307603  0.012948199 -0.011280314
## treat.fall_0_33     -0.21728951  0.021675314 -0.132101860
## treat.spring_0_33    0.19294270  0.009554993  0.106925041
## treat.spring_1_33    0.13554003 -0.036457122  0.090664352
## water_cap           -0.11958363  0.036606723 -0.054993642
## cover_non_seeded    -0.13121047  0.023002927 -0.049861600
## cover_seeded_grass   0.14341853 -0.015922400  0.005796685
```

RLQ Plots

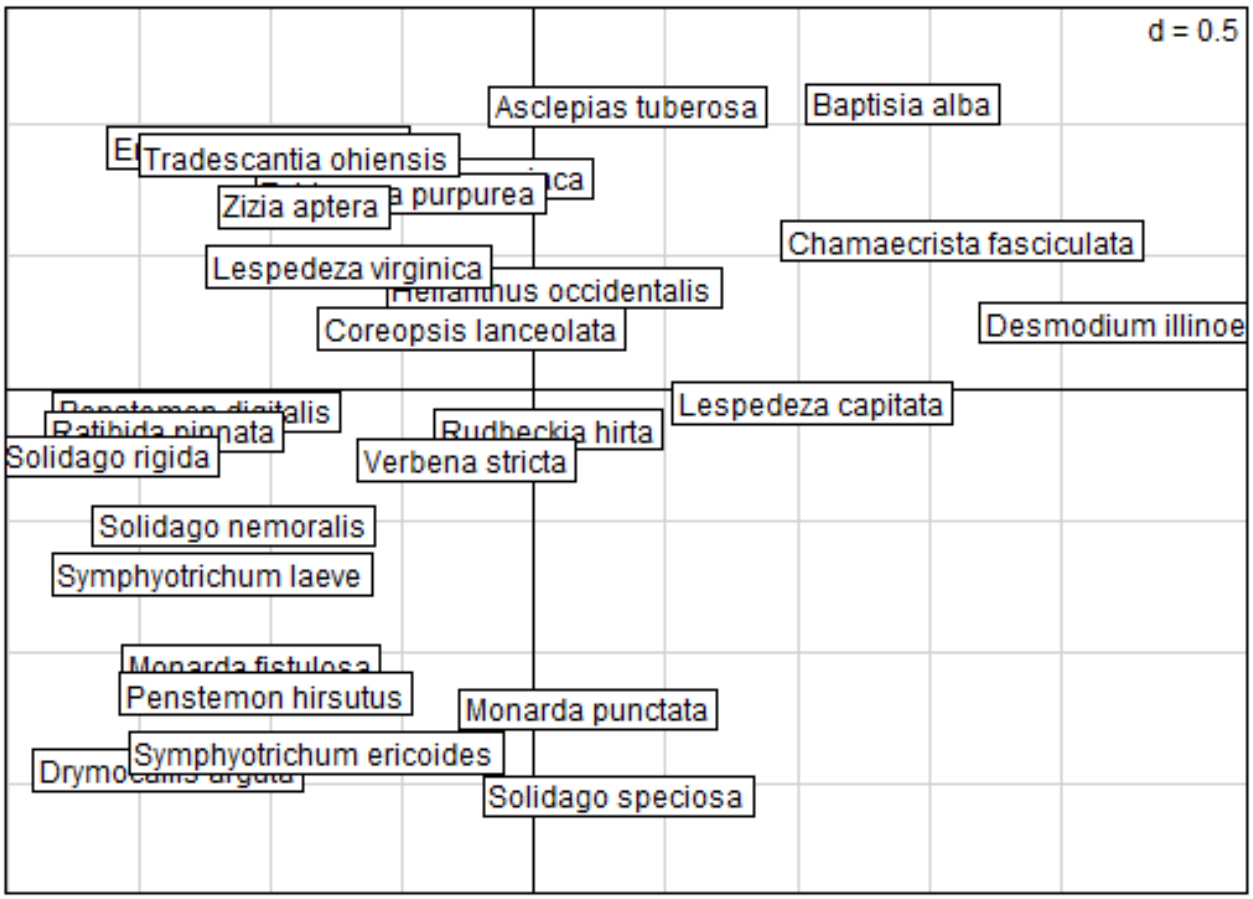
```
plot(m_rlq)
```



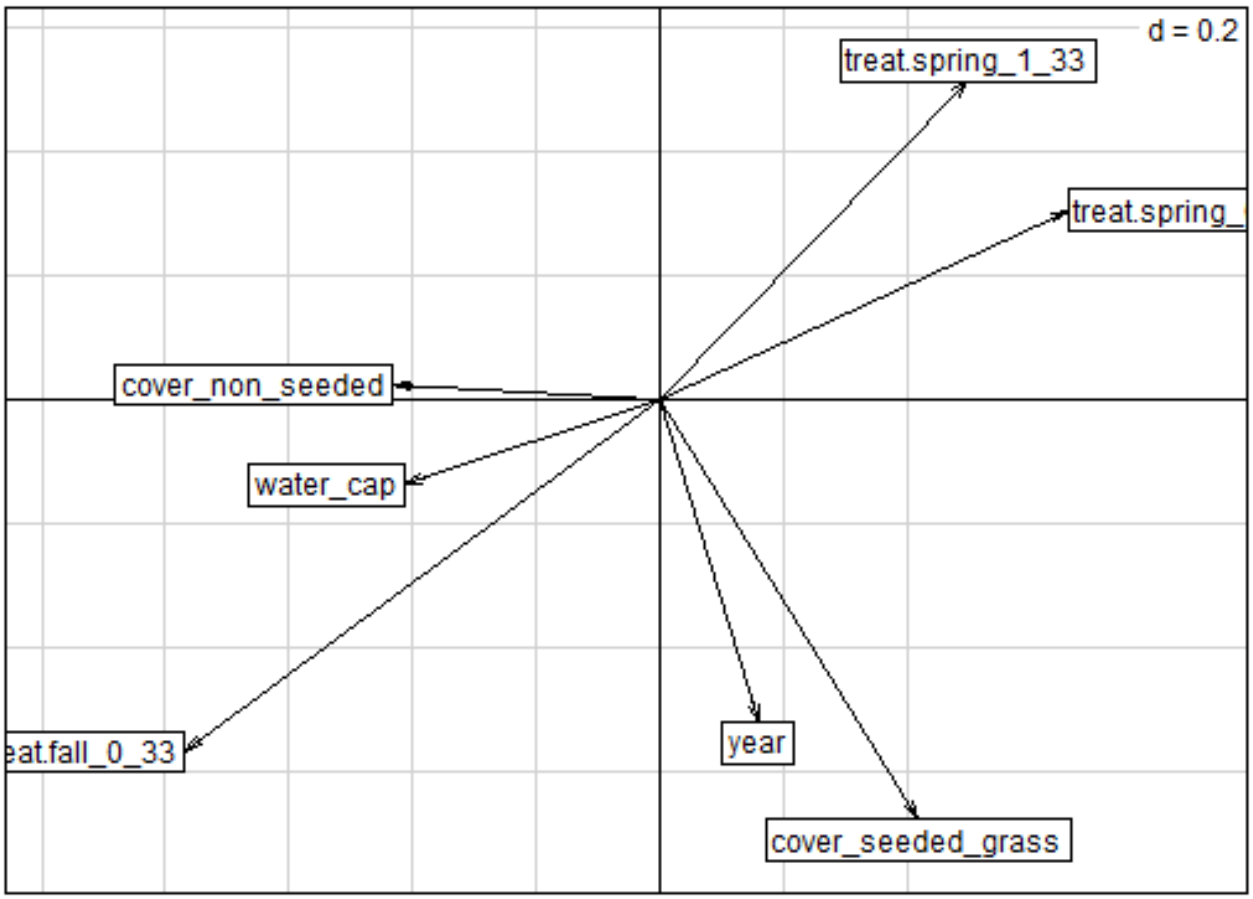

```
s.label(m_rlq$lR)
```



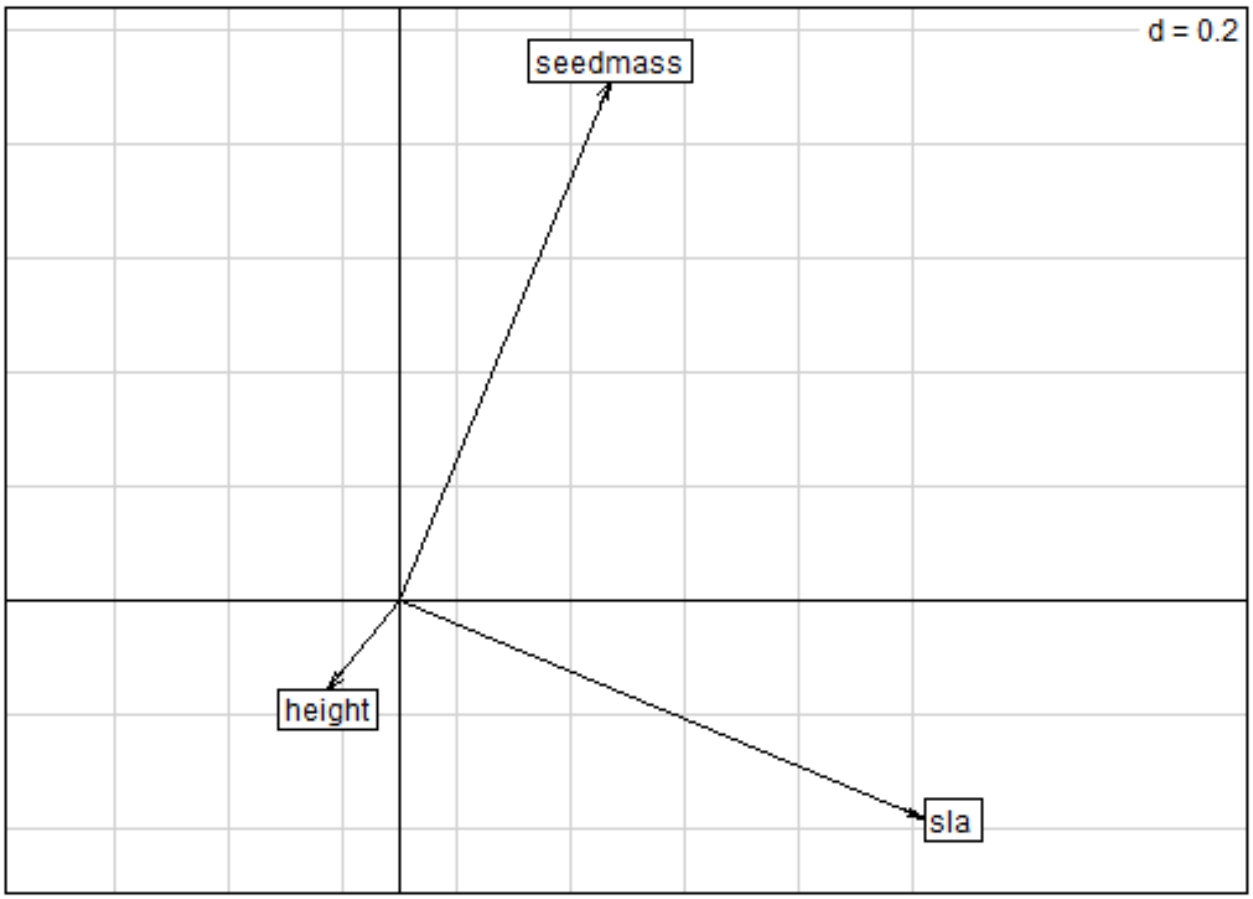

```
s.label(m_rlq$lQ)
```




```
s.arrow(m_rlq$l1)
```




```
s.arrow(m_rlq$c1)
```



Fourth-corner analysis

```
m_4th <- fourthcorner(
  tabR = sites,
  tabL = species,
  tabQ = traits,
  modeltype = 6,
  p.adjust.method.G = "none",
  p.adjust.method.D = "none",
  nrepet = 49999
)
m_4th # non-adjusted p-values
```

```
## Fourth-corner Statistics
## ------------------------
## Permutation method  Comb. 2 and 4  ( 49999  permutations)
##
## Adjustment method for multiple comparisons:   none
## call:  fourthcorner(tabR = sites, tabL = species, tabQ = traits, modeltype = 6,      nrepet = 49999,
##
## ---
##
##                              Test   Stat          Obs     Std.Obs     Alter
## 1                     year / sla      r  0.063076027  0.95985429 two-sided
## 2          treat.fall_0_33 / sla Homog.  0.331119500 -0.27728761      less
```

```
## 3        treat.spring_0_33 / sla Homog.  0.276597021  0.90679950      less
## 4        treat.spring_1_33 / sla Homog.  0.356835419  1.72638707      less
## 5                water_cap / sla      r -0.119583625 -2.82040627 two-sided
## 6         cover_non_seeded / sla      r -0.131210469 -2.33211019 two-sided
## 7       cover_seeded_grass / sla      r  0.143418534  2.17957998 two-sided
## 8                  year / height      r  0.012948199  0.19841281 two-sided
## 9       treat.fall_0_33 / height Homog.  0.563056753  6.90948464      less
## 10    treat.spring_0_33 / height Homog.  0.162891259 -3.46938174      less
## 11    treat.spring_1_33 / height Homog.  0.273403783 -1.63226735      less
## 12            water_cap / height      r  0.036606723  0.84631596 two-sided
## 13     cover_non_seeded / height      r  0.023002927  0.39850971 two-sided
## 14   cover_seeded_grass / height      r -0.015922400 -0.23517284 two-sided
## 15               year / seedmass      r -0.011280314 -0.17120168 two-sided
## 16    treat.fall_0_33 / seedmass Homog.  0.379014195  2.01454659      less
## 17  treat.spring_0_33 / seedmass Homog.  0.287549536  1.57197306      less
## 18  treat.spring_1_33 / seedmass Homog.  0.320469576  0.19495782      less
## 19          water_cap / seedmass      r -0.054993642 -1.29587626 two-sided
## 20   cover_non_seeded / seedmass      r -0.049861600 -0.87819573 two-sided
## 21 cover_seeded_grass / seedmass      r  0.005796685  0.08605095 two-sided
##     Pvalue
## 1    0.352
## 2   0.3899
## 3  0.80954
## 4  0.95784
## 5   0.0023  **
## 6  0.01466   *
## 7  0.02452   *
## 8  0.84772
## 9        1
## 10 0.00046 ***
## 11  0.0496   *
## 12 0.40806
## 13 0.70194
## 14 0.81874
## 15 0.87118
## 16  0.9782
## 17 0.94948
## 18 0.57194
## 19 0.20234
## 20 0.39562
## 21   0.933
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

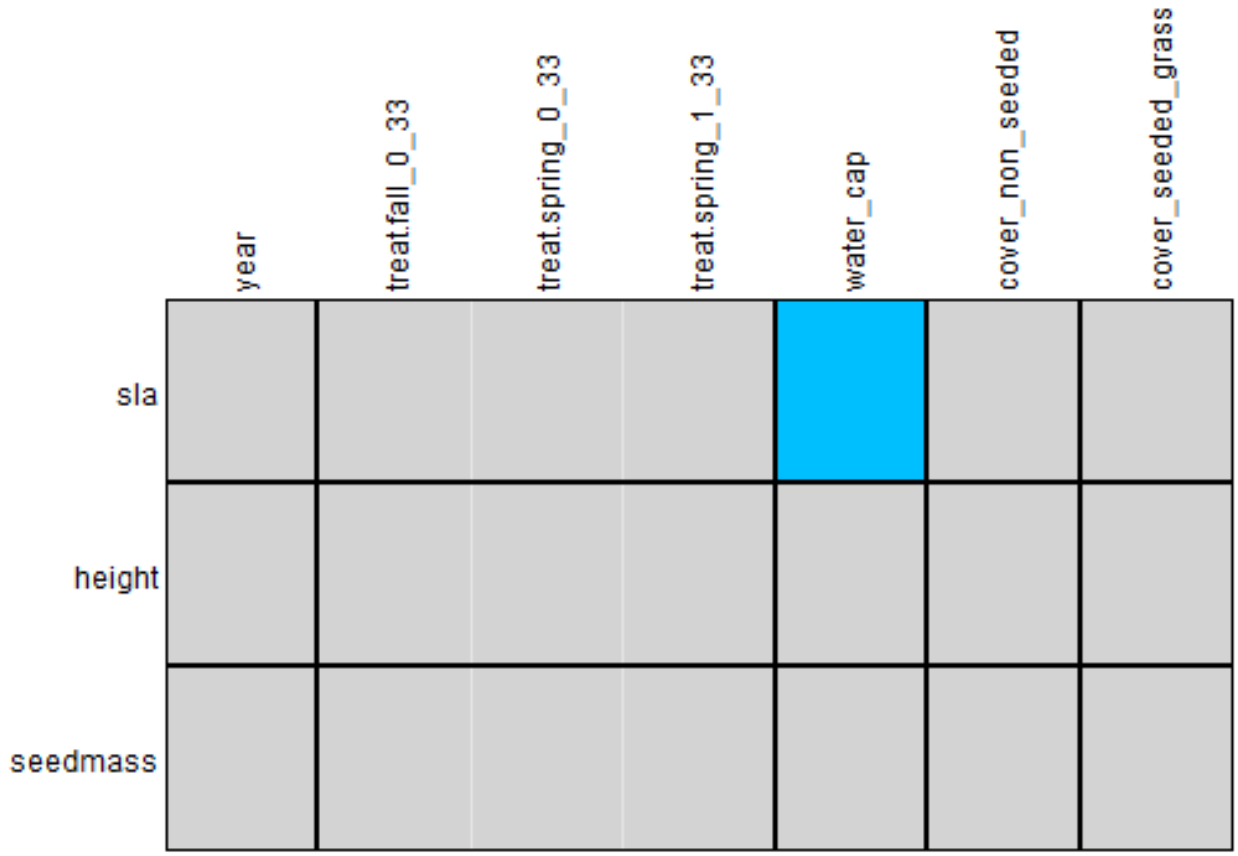
```r
m_4th_adj <- p.adjust.4thcorner(
  m_4th,
  p.adjust.method.G = "fdr",
  p.adjust.method.D = "fdr",
  p.adjust.D = "global"
)
m_4th_adj
```

```
## Fourth-corner Statistics
## ------------------------
```

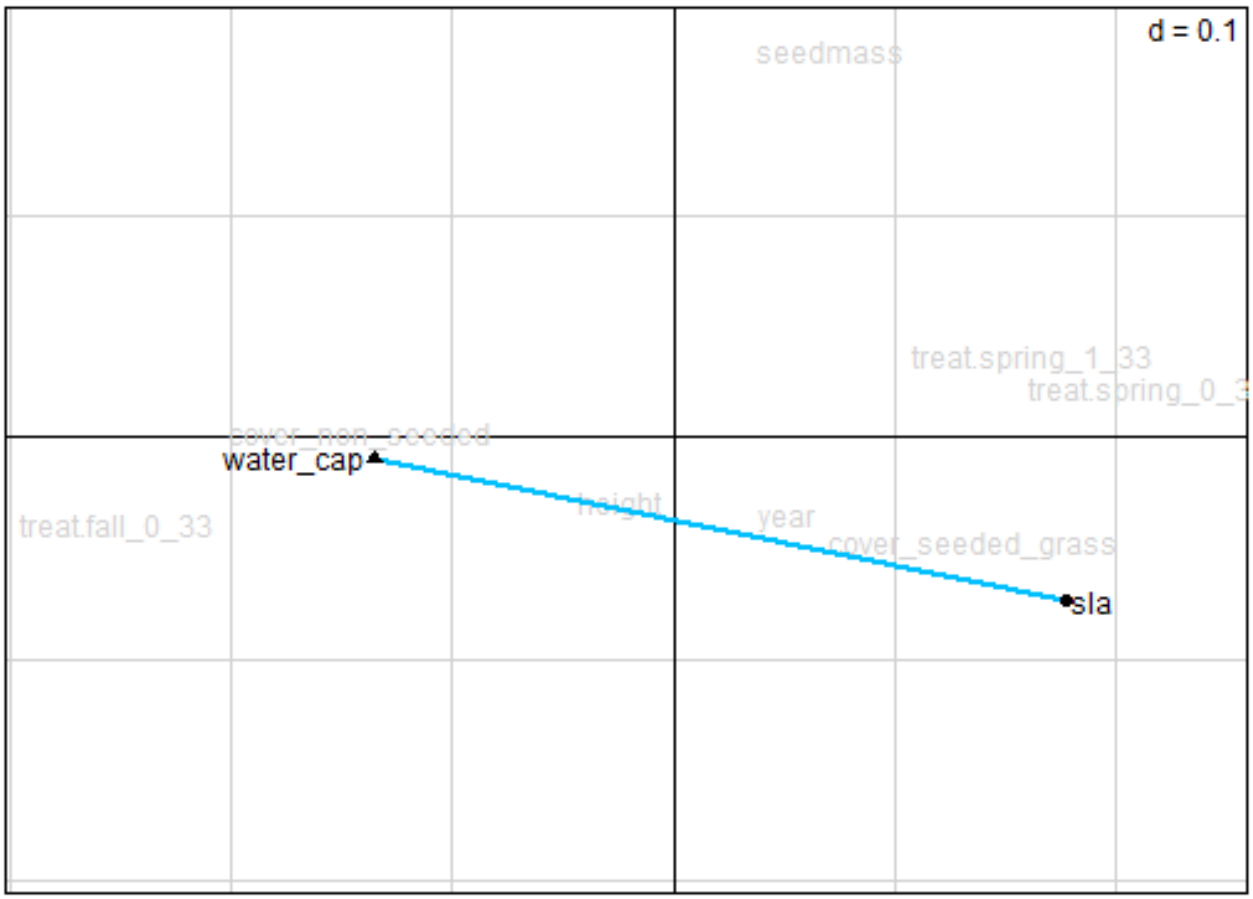
```
## Permutation method  Comb. 2 and 4  ( 49999  permutations)
##
## Adjustment method for multiple comparisons:   fdr
## call:  fourthcorner(tabR = sites, tabL = species, tabQ = traits, modeltype = 6,      nrepet = 49999,
##
## ---
##
##                              Test   Stat          Obs     Std.Obs     Alter
## 1                      year / sla      r  0.063076027  0.95985429 two-sided
## 2          treat.fall_0_33 / sla Homog.  0.331119500 -0.27728761      less
## 3        treat.spring_0_33 / sla Homog.  0.276597021  0.90679950      less
## 4        treat.spring_1_33 / sla Homog.  0.356835419  1.72638707      less
## 5                water_cap / sla      r -0.119583625 -2.82040627 two-sided
## 6         cover_non_seeded / sla      r -0.131210469 -2.33211019 two-sided
## 7       cover_seeded_grass / sla      r  0.143418534  2.17957998 two-sided
## 8                  year / height      r  0.012948199  0.19841281 two-sided
## 9       treat.fall_0_33 / height Homog.  0.563056753  6.90948464      less
## 10    treat.spring_0_33 / height Homog.  0.162891259 -3.46938174      less
## 11    treat.spring_1_33 / height Homog.  0.273403783 -1.63226735      less
## 12            water_cap / height      r  0.036606723  0.84631596 two-sided
## 13     cover_non_seeded / height      r  0.023002927  0.39850971 two-sided
## 14   cover_seeded_grass / height      r -0.015922400 -0.23517284 two-sided
## 15               year / seedmass      r -0.011280314 -0.17120168 two-sided
## 16    treat.fall_0_33 / seedmass Homog.  0.379014195  2.01454659      less
## 17  treat.spring_0_33 / seedmass Homog.  0.287549536  1.57197306      less
## 18  treat.spring_1_33 / seedmass Homog.  0.320469576  0.19495782      less
## 19          water_cap / seedmass      r -0.054993642 -1.29587626 two-sided
## 20   cover_non_seeded / seedmass      r -0.049861600 -0.87819573 two-sided
## 21 cover_seeded_grass / seedmass      r  0.005796685  0.08605095 two-sided
##     Pvalue Pvalue.adj
## 1    0.352   0.856926
## 2   0.3899   0.856926
## 3  0.80954          1
## 4  0.95784          1
## 5   0.0023    0.02415  *
## 6  0.01466    0.10262
## 7  0.02452    0.12873
## 8  0.84772          1
## 9        1          1
## 10 0.00046    0.00966 **
## 11  0.0496    0.20832
## 12 0.40806   0.856926
## 13 0.70194          1
## 14 0.81874          1
## 15 0.87118          1
## 16  0.9782          1
## 17 0.94948          1
## 18 0.57194          1
## 19 0.20234    0.70819
## 20 0.39562   0.856926
## 21   0.933          1
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Plot

```r
plot(m_4th_adj, alpha = 0.05, stat = "D2")
```




```r
plot(m_4th_adj,
  x.rlq = m_rlq,
  alpha = 0.05,
  stat = "D2",
  type = "biplot"
)
```



Save

Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] ade4_1.7-23     lubridate_1.9.4 forcats_1.0.0   stringr_1.5.1
```

```
##  [5] dplyr_1.1.4     purrr_1.1.0     readr_2.1.5     tidyr_1.3.1
##  [9] tibble_3.3.0    ggplot2_3.5.2   tidyverse_2.0.0 here_1.0.1
## [13] tinytex_0.57
##
## loaded via a namespace (and not attached):
##  [1] bit_4.6.0          gtable_0.3.6       crayon_1.5.3       compiler_4.5.1
##  [5] Rcpp_1.1.0         tidyselect_1.2.1   parallel_4.5.1     scales_1.4.0
##  [9] yaml_2.3.10        fastmap_1.2.0      R6_2.6.1           generics_0.1.4
## [13] knitr_1.50         MASS_7.3-65        rprojroot_2.1.1    pillar_1.11.0
## [17] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.6        utf8_1.2.6
## [21] stringi_1.8.7      xfun_0.52          bit64_4.6.0-1      timechange_0.3.0
## [25] cli_3.6.5          withr_3.0.2        magrittr_2.0.3     digest_0.6.37
## [29] grid_4.5.1         vroom_1.6.5        rstudioapi_0.17.1  hms_1.1.3
## [33] lifecycle_1.0.4    vctrs_0.6.5        evaluate_1.0.4     glue_1.8.0
## [37] farver_2.1.2       rmarkdown_2.29     tools_4.5.1        pkgconfig_2.0.3
## [41] htmltools_0.5.8.1
```